

Technology #03: Catalytic Contact Time Lock for Amyloid Beta Clearance

Technology Domain: Enzymatic Catalysis + Mass Transport

Mechanism: Catheter with immobilized neprilysin (NEP) enzyme on inner lumen wall; CSF flows past, A β 42 in CSF contacts enzyme for sufficient time (contact-time lock) to undergo catalytic cleavage

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Enzyme half-life in CSF environment is less than 30 days (insufficient for chronic implant)

Next Experiment: Recombinant NEP enzyme lot

Buyer Receives: Conceptual design + governing equation (Michaelis-Menten + Damköhler); Mechanism description (immobilized NEP on catheter lumen); Bench test protocol (clearance per pass, stability); External precedent catalog (NEP kinetics, immobilization chemistry); This engineering dossier with failure analysis + build plan

Buyer Must Build: Recombinant NEP production process (GMP-grade); Coating process specification + QC; Final catheter geometry (length, diameter, coating density); Manufacturing process for enzyme-active catheter (aseptic assembly); Regulatory submission (likely Class III PMA — no predicate for enzymatic catheter); Clinical validation (CSF A β 42 reduction + clinical benefit endpoint); Long-term stability and reliability data

Transfer Posture: Sponsored Validation

Target Buyer: Biotech company with enzyme therapy program; neuroshunt manufacturer

CONFIDENTIAL. See full engineering dossier for details.